

Asymptotic Fisher information for Poisson counts driven by a stationary Gaussian process

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This memo rigorously proves Eq. (3.17) in Ng *et al.*, Phys. Rev. A **93**, 042121 (2016) for the asymptotic Fisher information of spectral photon counting. Let a real zero-mean stationary Gaussian process have spectral density R_θ , and let the observations be conditionally independent Poisson random variables whose means are the squared moduli of the positive-frequency discrete Fourier coefficients of a length- T sample. Under Wiener-algebra regularity and strict positivity of the spectrum, we prove

$$\lim_{T \rightarrow \infty} \frac{I_T(\theta)}{T} = \frac{1}{2\pi} \int_0^\pi \frac{[\partial_\theta R_\theta(\omega)]^2}{R_\theta(\omega)[1 + R_\theta(\omega)]} d\omega.$$

ChatGPT 5.6 Sol produced the proof, following a suggestion by Tsang. Claude Opus 5.5 checked the proof and proposed minor improvements.

I. INTRODUCTION

The goal of this work is to rigorously prove Eq. (3.17) in Ref. [1] for the asymptotic Fisher information of spectral photon counting, after a zero-mean stationary Gaussian process with a parameter-dependent spectral density modulates a coherent optical beam.

Ref. [1] derives the formula by showing that the photon counts tend to a set of independent Bose–Einstein (geometric) random variables in the long-time limit and then computing the Fisher information of the Bose–Einstein variables. However, ChatGPT complained that this approach is not rigorous, as one should instead take the long-time limit of the Fisher information to determine the asymptotic Cramér–Rao bound; taking the long-time limit prematurely need not give the correct result.

Following Refs. [1, 2], Tsang suggested to ChatGPT the general strategy of sandwiching the Fisher information between a lower bound due to Stein, Mezghani, and Nossek [3] and an upper bound due to Zamir [4], as he suspected that the two bounds, once optimized, should give the same asymptotic formula. ChatGPT took the suggestion and eventually produced the proof below, after a few rounds of encouragement by Tsang. The proof was then bounced between Claude and ChatGPT for checking and improvements. Both eventually claimed the proof to be correct, although Tsang has not checked it independently.

II. MODEL AND ASSUMPTIONS

Identify $\mathbb{T}$ with $[-\pi, \pi)$ and let $A(\mathbb{T})$ denote the Wiener algebra

$$A(\mathbb{T}) = \left\{ a(\omega) = \sum_{k \in \mathbb{Z}} a_k e^{-ik\omega} : \sum_{k \in \mathbb{Z}} |a_k| < \infty \right\}, \quad \|a\|_A = \sum_k |a_k|.$$

Fix a parameter value θ_0 . Assume that on an open interval Θ containing θ_0 :

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(A1) $\theta \mapsto R_\theta$ is C^1 as a map $\Theta \rightarrow A(\mathbb{T})$;

(A2) R_θ is real and even, and there are constants $0 < m < M < \infty$ such that

$$m \leq R_\theta(\omega) \leq M, \quad \theta \in \Theta, \omega \in \mathbb{T}.$$

No separate likelihood-regularity assumption is imposed: the finite-dimensional Gaussian and Gaussian–Poisson models used below are shown in Appendix A to possess the score identities and finite Fisher information required by the proof.

Let $Z = (Z_t)_{t \in \mathbb{Z}}$ be a real, zero-mean stationary Gaussian process with spectral density R_θ . Thus

$$C_\theta(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_\theta(\omega) e^{ik\omega} d\omega, \quad R_\theta(\omega) = \sum_{k \in \mathbb{Z}} C_\theta(k) e^{-ik\omega}.$$

For $T \geq 3$ let $\omega_j = 2\pi j/T$ and let F_T be the unitary DFT matrix

$$(F_T)_{jt} = T^{-1/2} e^{-i\omega_j t}, \quad j = 0, \dots, T-1, \quad t = 1, \dots, T.$$

For $Z^{(T)} = (Z_1, \dots, Z_T)^\top$ put $Y_T = F_T Z^{(T)}$, that is,

$$Y_{T,j} = \frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t e^{-i\omega_j t}, \quad j = 0, \dots, T-1,$$

and let $J_T = \{j : 0 < \omega_j < \pi\}$. Conditional on $Z^{(T)}$, the observations are independent and

$$N_{T,j} \mid Z^{(T)} \sim \text{Pois}(|Y_{T,j}|^2), \quad j \in J_T. \quad (2.1)$$

Write $I_T(\theta)$ for the Fisher information in $N_T = (N_{T,j})_{j \in J_T}$.

Theorem 1. *Under (A1)–(A2),*

$$\lim_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} = \frac{1}{2\pi} \int_0^\pi \frac{\dot{R}(\omega)^2}{R(\omega)[1 + R(\omega)]} d\omega, \quad (2.2)$$

where $R = R_{\theta_0}$ and $\dot{R} = \partial_\theta R_\theta|_{\theta_0}$.

Theorem 1 leads to Eq. (3.17) in Ref. [1] after some trivial rescaling.

We write $\mathcal{I}$ for the right-hand side of (2.2). Throughout, a dot denotes ∂_θ , and quantities written without a θ subscript are evaluated at θ_0 . Theorem 1 follows from Proposition 1 (Section IV) and Proposition 2 (Section V).

III. TOEPLITZ–CIRCULANT LEMMAS

For $a \in A(\mathbb{T})$ define the Toeplitz matrix $T_T(a) = (a_{r-s})_{r,s=1}^T$, so that $\text{Cov}(Z^{(T)}) = T_T(R_\theta)$. The map $a \mapsto T_T(a)$ is linear, $T_T(1) = I$, and $T_T(a)$ is Hermitian when a is real. For $v \in \mathbb{C}^T$,

$$v^* T_T(a) v = \frac{1}{2\pi} \int_{-\pi}^{\pi} a(\omega) \left| \sum_{r=1}^T v_r e^{-ir\omega} \right|^2 d\omega, \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{r=1}^T v_r e^{-ir\omega} \right|^2 d\omega = \|v\|^2, \quad (3.1)$$

so $(\min a)I \leq T_T(a) \leq (\max a)I$ for real a . In particular, $mI \leq T_T(R_\theta) \leq MI$ by (A2).

For a bounded function a on $\mathbb{T}$ define the sampled circulant

$$\mathcal{C}_T(a) = F_T^* \text{diag}(a(\omega_0), \dots, a(\omega_{T-1})) F_T.$$

Sampled circulants are simultaneously diagonalized by F_T , so $\mathcal{C}_T(a)\mathcal{C}_T(b) = \mathcal{C}_T(ab)$, $\mathcal{C}_T(a)^* = \mathcal{C}_T(\bar{a})$, $\|\mathcal{C}_T(a)\|_{\text{op}} = \max_j |a(\omega_j)|$, and $f(\mathcal{C}_T(a)) = \mathcal{C}_T(f \circ a)$ for every function f defined on the values $a(\omega_j)$; in particular $\mathcal{C}_T(a)^{-1} = \mathcal{C}_T(1/a)$ if a does not vanish.

For $T \times T$ matrices set $\|M\|_{F,T} = T^{-1/2}\|M\|_F$. This norm is unitarily invariant, and for all matrices M, M' and every orthogonal projection P ,

$$\begin{aligned} \|M\|_{F,T} &\leq \|M\|_{\text{op}}, & \frac{|\text{tr}(PM)|}{T} &\leq \|M\|_{F,T}, \\ \|MM'\|_{F,T} &\leq \min\{\|M\|_{F,T}\|M'\|_{\text{op}}, \|M\|_{\text{op}}\|M'\|_{F,T}\}. \end{aligned} \quad (3.2)$$

(The trace bound follows from $|\text{tr}(PM)| \leq \|P\|_F\|M\|_F$ and $\|P\|_F \leq T^{1/2}$.) We write $M_T \sim \widetilde{M}_T$ if $\|M_T - \widetilde{M}_T\|_{F,T} \rightarrow 0$ and both sequences have uniformly bounded operator norm. Finally, for continuous g on $\mathbb{T}$ the Riemann-sum theorem gives

$$\frac{1}{T} \sum_{j=0}^{T-1} g(\omega_j) \rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\omega) d\omega, \quad \frac{1}{T} \sum_{j \in J_T} g(\omega_j) \rightarrow \frac{1}{2\pi} \int_0^{\pi} g(\omega) d\omega. \quad (3.3)$$

Lemma 1 (Toeplitz–circulant equivalence). *If $a \in A(\mathbb{T})$, then $T_T(a) \sim \mathcal{C}_T(a)$. Moreover, both $\|T_T(a)\|_{\text{op}}$ and $\|\mathcal{C}_T(a)\|_{\text{op}}$ are at most $\|a\|_A$.*

Proof. The bound for $T_T(a)$ is the ℓ^1 convolution bound, and $\|\mathcal{C}_T(a)\|_{\text{op}} = \max_j |a(\omega_j)| \leq \|a\|_A$. Since $\sum_k |a_k| < \infty$,

$$(\mathcal{C}_T(a))_{rs} = \frac{1}{T} \sum_{j=0}^{T-1} a(\omega_j) e^{i\omega_j(r-s)} = \sum_{\ell \in \mathbb{Z}} a_{r-s+\ell T}.$$

First suppose a is a trigonometric polynomial of degree d and $T > 2d$. Then $T_T(a)$ and $\mathcal{C}_T(a)$ agree except at the $d(d+1)$ entries with $|r-s| \geq T-d$, where they differ by at most $\max_k |a_k|$. Hence $\|T_T(a) - \mathcal{C}_T(a)\|_F = O_d(1)$, and the normalized Frobenius norm tends to zero.

For general $a \in A(\mathbb{T})$, choose a trigonometric polynomial b with $\|a - b\|_A < \varepsilon$. By (3.2),

$$\|T_T(a - b)\|_{F,T} \leq \|T_T(a - b)\|_{\text{op}} \leq \varepsilon,$$

and similarly for $\mathcal{C}_T(a - b)$. Thus $\limsup_T \|T_T(a) - \mathcal{C}_T(a)\|_{F,T} \leq 2\varepsilon$, and letting $\varepsilon \downarrow 0$ proves the claim. $\square$

Lemma 2 (Stability). *Suppose $M_T \sim \widetilde{M}_T$ and $M'_T \sim \widetilde{M}'_T$. Then $M_T + M'_T \sim \widetilde{M}_T + \widetilde{M}'_T$, $M_T^* \sim \widetilde{M}_T^*$, and $M_T M'_T \sim \widetilde{M}_T \widetilde{M}'_T$. If, in addition, M_T and $\widetilde{M}_T$ are invertible and their inverses have uniformly bounded operator norm, then $M_T^{-1} \sim \widetilde{M}_T^{-1}$.*

Proof. Sums and adjoints are immediate, and the operator norms of products and inverses are bounded by submultiplicativity. For products write

$$M_T M'_T - \widetilde{M}_T \widetilde{M}'_T = (M_T - \widetilde{M}_T) M'_T + \widetilde{M}_T (M'_T - \widetilde{M}'_T),$$

and bound the first term by $\|M_T - \widetilde{M}_T\|_{F,T} \|M'_T\|_{\text{op}}$ and the second by $\|\widetilde{M}_T\|_{\text{op}} \|M'_T - \widetilde{M}'_T\|_{F,T}$, using both inequalities in (3.2). For inverses use

$$M_T^{-1} - \widetilde{M}_T^{-1} = M_T^{-1} (\widetilde{M}_T - M_T) \widetilde{M}_T^{-1}. \quad \square$$

Lemma 3 (Functional calculus). *Let $\theta \mapsto a_\theta$ be C^1 from Θ into $A(\mathbb{T})$, with a_θ real valued and $a_\theta(\omega) \in [\alpha, \beta] \subset (0, \infty)$ for all $\theta \in \Theta$ and $\omega \in \mathbb{T}$. Let f be analytic on an open set $\Omega \subset \mathbb{C}$ containing $[\alpha, \beta]$, and write $a = a_{\theta_0}$ and $\dot{a} = \partial_\theta a_\theta|_{\theta_0}$. Then:*

(i) $f(T_T(a)) \sim f(\mathcal{C}_T(a)) = \mathcal{C}_T(f \circ a)$;

(ii) for each T , $\theta \mapsto f(T_T(a_\theta))$ is C^1 in operator norm, and

$$\partial_\theta f(T_T(a_\theta))|_{\theta_0} \sim \partial_\theta f(\mathcal{C}_T(a_\theta))|_{\theta_0} = \mathcal{C}_T(f'(a) \dot{a}).$$

Proof. By (3.1), $\alpha I \leq T_T(a_\theta) \leq \beta I$, and the eigenvalues $a_\theta(\omega_j)$ of $\mathcal{C}_T(a_\theta)$ also lie in $[\alpha, \beta]$. Fix a contour $\gamma \subset \Omega \setminus [\alpha, \beta]$ winding once around $[\alpha, \beta]$, and let $\delta = \text{dist}(\gamma, [\alpha, \beta]) > 0$. Both matrices are normal with spectrum in $[\alpha, \beta]$, so their resolvents have operator norm at most δ^{-1} on γ . Write $M_T = T_T(a)$, $\widetilde{M}_T = \mathcal{C}_T(a)$, $\mathcal{R}(z) = (zI - M_T)^{-1}$ and $\widetilde{\mathcal{R}}(z) = (zI - \widetilde{M}_T)^{-1}$. The resolvent identity $\mathcal{R} - \widetilde{\mathcal{R}} = \mathcal{R}(M_T - \widetilde{M}_T)\widetilde{\mathcal{R}}$, (3.2) and Lemma 1 give

$$\sup_{z \in \gamma} \|\mathcal{R}(z) - \widetilde{\mathcal{R}}(z)\|_{F,T} \leq \delta^{-2} \|M_T - \widetilde{M}_T\|_{F,T} \rightarrow 0. \quad (3.4)$$

(i) Integrating (3.4) against f in $f(M) = \frac{1}{2\pi i} \oint_\gamma f(z)(zI - M)^{-1} dz$ gives $\|f(M_T) - f(\widetilde{M}_T)\|_{F,T} \rightarrow 0$, and both matrices have operator norm at most $|\gamma| \sup_\gamma |f|/(2\pi\delta)$, where $|\gamma|$ is the length of γ . Diagonalization gives $f(\widetilde{M}_T) = \mathcal{C}_T(f \circ a)$.

(ii) Since T_T is linear and $\|T_T(b)\|_{\text{op}} \leq \|b\|_A$, the map $\theta \mapsto T_T(a_\theta)$ is C^1 in operator norm with derivative $T_T(\dot{a}_\theta)$; likewise $\theta \mapsto \mathcal{C}_T(a_\theta)$ has derivative $\mathcal{C}_T(\dot{a}_\theta)$. On the open set of matrices with spectrum inside γ , $M \mapsto f(M)$ is continuously Fréchet differentiable, with

$$Df(M)[\Delta] = \frac{1}{2\pi i} \oint_\gamma f(z)(zI - M)^{-1} \Delta (zI - M)^{-1} dz.$$

By the chain rule, $\theta \mapsto f(T_T(a_\theta))$ is C^1 , $\partial_\theta f(T_T(a_\theta))|_{\theta_0} = Df(M_T)[T_T(\dot{a})]$, and $\partial_\theta f(\mathcal{C}_T(a_\theta))|_{\theta_0} = Df(\widetilde{M}_T)[\mathcal{C}_T(\dot{a})]$. The difference of the two integrands is

$$(\mathcal{R} - \widetilde{\mathcal{R}}) T_T(\dot{a}) \mathcal{R} + \widetilde{\mathcal{R}} \{T_T(\dot{a}) - \mathcal{C}_T(\dot{a})\} \mathcal{R} + \widetilde{\mathcal{R}} \mathcal{C}_T(\dot{a}) (\mathcal{R} - \widetilde{\mathcal{R}}),$$

which is $o(1)$ in $\|\cdot\|_{F,T}$, uniformly on γ , by (3.2), (3.4), and Lemma 1 applied to $\dot{a}$. Both derivatives have operator norm at most $|\gamma| \sup_\gamma |f| \|\dot{a}\|_A/(2\pi\delta^2)$. Finally, $f(\mathcal{C}_T(a_\theta)) = F_T^* \text{diag}(f(a_\theta(\omega_j))) F_T$ for all θ , and point evaluation is continuous on $A(\mathbb{T})$, so differentiating entrywise gives $\mathcal{C}_T(f'(a)\dot{a})$. $\square$

IV. LOWER BOUND

Set

$$\Gamma_T = T_T(R), \quad A_T = F_T \Gamma_T F_T^*, \quad B_T = F_T \Gamma_T F_T^\top.$$

Since $Y_T = F_T Z^{(T)}$,

$$(A_T)_{jk} = \mathbb{E}(Y_{T,j} \overline{Y_{T,k}}), \quad (B_T)_{jk} = \mathbb{E}(Y_{T,j} Y_{T,k}).$$

Write $s_{T,j} = (A_T)_{jj} = \mathbb{E}|Y_{T,j}|^2$, and note that $0 < s_{T,j} \leq M$ because $mI \leq \Gamma_T \leq MI$.

The real and imaginary parts of the $Y_{T,j}$ are jointly Gaussian with mean zero, so Isserlis' theorem gives

$$\text{Cov}(|Y_{T,j}|^2, |Y_{T,k}|^2) = |(A_T)_{jk}|^2 + |(B_T)_{jk}|^2.$$

Using conditional independence and the variance formula for a Poisson variable,

$$\text{Cov}(N_{T,j}, N_{T,k}) = \delta_{jk} s_{T,j} + |(A_T)_{jk}|^2 + |(B_T)_{jk}|^2, \quad j, k \in J_T. \quad (4.1)$$

Define

$$w(\omega) = \frac{\dot{R}(\omega)}{R(\omega)[1 + R(\omega)]}, \quad w_j = w(\omega_j).$$

By (A1)–(A2), w is bounded and continuous. At the fixed parameter value θ_0 , define the scalar statistic

$$Q_T = \sum_{j \in J_T} w_j N_{T,j}. \quad (4.2)$$

The weights are held fixed when differentiating in θ .

Proposition 1 (Lower bound). *Under (A1)–(A2), $\liminf_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \geq \mathcal{I}$.*

Proof. By Lemma A.1 in Appendix A, $\partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0} = \mathbb{E}_{\theta_0}[(Q_T - \mathbb{E}_{\theta_0} Q_T) U_{N, \theta_0}]$, where $U_{N, \theta}$ is the score of N_T . The Cauchy–Schwarz inequality therefore yields the scalar information inequality

$$I_T(\theta_0) \geq \frac{[\partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0}]^2}{\text{Var}_{\theta_0}(Q_T)} \quad (4.3)$$

whenever $\text{Var}_{\theta_0}(Q_T) > 0$.

Numerator. We have $\mathbb{E}_\theta Q_T = \sum_{j \in J_T} w_j s_{T,j}(\theta)$ with $s_{T,j}(\theta) = (F_T T_T(R_\theta) F_T^*)_{jj}$, so

$$\partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0} = \sum_{j \in J_T} w_j \dot{s}_{T,j}, \quad \dot{s}_{T,j} = (F_T T_T(\dot{R}) F_T^*)_{jj}.$$

By unitary invariance and Lemma 1 applied to $\dot{R}$,

$$\frac{1}{T} \sum_{j=0}^{T-1} |\dot{s}_{T,j} - \dot{R}(\omega_j)|^2 \leq \|T_T(\dot{R}) - \mathcal{C}_T(\dot{R})\|_{F,T}^2 \rightarrow 0.$$

Since w is bounded, the Cauchy–Schwarz inequality and (3.3) imply

$$\frac{1}{T} \partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0} \longrightarrow \frac{1}{2\pi} \int_0^\pi w(\omega) \dot{R}(\omega) d\omega = \mathcal{I}. \quad (4.4)$$

Variance. By Lemma 1 and unitary invariance,

$$\sum_{j,k=0}^{T-1} |(A_T)_{jk} - \delta_{jk} R(\omega_j)|^2 = \|\Gamma_T - \mathcal{C}_T(R)\|_F^2 = o(T). \quad (4.5)$$

Hence

$$\sum_{j \neq k} |(A_T)_{jk}|^2 = o(T), \quad \sum_j |s_{T,j} - R(\omega_j)|^2 = o(T). \quad (4.6)$$

Let Π_T be the permutation matrix of $j \mapsto -j \pmod{T}$. A direct computation gives $F_T F_T^\top = \Pi_T$, hence

$$F_T \mathcal{C}_T(R) F_T^\top = \text{diag}(R(\omega_j)) \Pi_T.$$

For $j, k \in J_T$ we have $2 \leq j + k \leq T - 1$, so $k \not\equiv -j \pmod{T}$, and the $J_T \times J_T$ block of this matrix vanishes. As $F_T^\top$ is unitary, Lemma 1 gives

$$\sum_{j,k \in J_T} |(B_T)_{jk}|^2 \leq \|\Gamma_T - \mathcal{C}_T(R)\|_F^2 = o(T). \quad (4.7)$$

By (4.1),

$$\text{Var}(Q_T) = \sum_{j \in J_T} w_j^2 (s_{T,j} + s_{T,j}^2) + \sum_{\substack{j,k \in J_T \\ j \neq k}} w_j w_k |(A_T)_{jk}|^2 + \sum_{j,k \in J_T} w_j w_k |(B_T)_{jk}|^2,$$

and by boundedness of w , (4.6) and (4.7), the last two sums are $o(T)$. Since $|s_{T,j}^2 - R(\omega_j)^2| \leq 2M|s_{T,j} - R(\omega_j)|$, the second relation in (4.6), the Cauchy–Schwarz inequality and (3.3) yield

$$\frac{1}{T} \text{Var}(Q_T) \longrightarrow \frac{1}{2\pi} \int_0^\pi w(\omega)^2 R(\omega) [1 + R(\omega)] d\omega = \mathcal{I}. \quad (4.8)$$

If $\mathcal{I} > 0$, then $\text{Var}(Q_T) > 0$ for large T , and (4.3), (4.4) and (4.8) imply

$$\liminf_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \geq \mathcal{I}. \quad (4.9)$$

If $\mathcal{I} = 0$, (4.9) follows from nonnegativity of Fisher information. $\square$

V. UPPER BOUND BY EXACT GAUSSIAN AUGMENTATION

Define, for every $\theta \in \Theta$,

$$S_\theta = 1 + R_\theta, \quad H_\theta = \frac{R_\theta}{1 + R_\theta}. \quad (5.1)$$

Then $R_\theta = S_\theta H_\theta$ and, since $T_T(1) = I$, $T_T(S_\theta) = I + T_T(R_\theta)$. Let

$$X^{(T)} \sim N(0, T_T(S_\theta))$$

and set

$$L_T(\theta) = T_T(R_\theta)^{1/2} T_T(S_\theta)^{-1/2}, \quad Z^{(T)} = L_T(\theta) X^{(T)}. \quad (5.2)$$

Both factors of $L_T(\theta)$ are functions of the same real symmetric matrix $T_T(R_\theta)$, so

$$L_T(\theta) = h(T_T(R_\theta)), \quad h(x) = \sqrt{\frac{x}{1+x}}. \quad (5.3)$$

In particular $L_T(\theta)$ is real symmetric, and $L_T(\theta) T_T(S_\theta) L_T(\theta)^\top = T_T(R_\theta)$ because $h(x)^2(1+x) = x$. Thus $Z^{(T)}$ has exactly the original Gaussian law, and if N_T is generated from $Z^{(T)}$ by (2.1), its marginal law is exactly the original law, for every T and every $\theta \in \Theta$. By Lemma A.2 in Appendix A, applied to (5.1),

$$I_T(\theta_0) \leq I_T^X + \mathbb{E} I_T^{N|X}(X^{(T)}), \quad (5.4)$$

where I_T^X is the Fisher information of $X^{(T)}$ and $I_T^{N|X}(x)$ is the conditional Fisher information of N_T given $X^{(T)} = x$, both at θ_0 .

Proposition 2 (Upper bound). *Under (A1)–(A2), $\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \mathcal{I}$.*

Proof. Gaussian term. For a zero-mean real Gaussian vector whose positive definite covariance G_θ is C^1 in θ , the Fisher information is $\frac{1}{2} \text{tr}(G^{-1} \dot{G} G^{-1} \dot{G})$. Therefore

$$I_T^X = \frac{1}{2} \text{tr}\{T_T(S)^{-1} T_T(\dot{S}) T_T(S)^{-1} T_T(\dot{S})\}.$$

By Lemma 1, $T_T(S) \sim \mathcal{C}_T(S)$ and $T_T(\dot{S}) \sim \mathcal{C}_T(\dot{S})$. Since $T_T(S) \geq (1+m)I$ and $\mathcal{C}_T(S) \geq (1+m)I$, Lemma 2 gives $T_T(S)^{-1} \sim \mathcal{C}_T(1/S)$, and then that the product inside the trace is asymptotically equivalent to $\mathcal{C}_T((\dot{S}/S)^2)$. By (3.2) with $P = I$, normalized traces of asymptotically equivalent matrices differ by $o(1)$, so (3.3) and evenness give

$$\lim_{T \rightarrow \infty} \frac{I_T^X}{T} = \frac{1}{4\pi} \int_{-\pi}^{\pi} \left(\frac{\dot{S}}{S} \right)^2 d\omega = \frac{1}{2\pi} \int_0^\pi (\partial_\theta \log S)^2 d\omega. \quad (5.5)$$

Conditional Poisson term. At θ_0 put

$$Y = F_T L_T X^{(T)}, \quad V = F_T \dot{L}_T X^{(T)},$$

where $\dot{L}_T = \partial_\theta L_T(\theta)|_{\theta_0}$. Conditional on $X^{(T)}$, the Poisson mean at frequency $j \in J_T$ is $\lambda_j = |Y_j|^2$, which is almost surely positive (see Lemma A.2), and, holding $X^{(T)}$ fixed,

$$\dot{\lambda}_j = 2 \text{Re}(\overline{Y_j} V_j).$$

The conditional Fisher information is therefore

$$I_T^{N|X}(X^{(T)}) = \sum_{j \in J_T} \frac{\dot{\lambda}_j^2}{\lambda_j} = 4 \sum_{j \in J_T} \frac{[\text{Re}(\overline{Y_j} V_j)]^2}{|Y_j|^2},$$

and since $[\text{Re}(\bar{y}v)]^2 \leq |y|^2 |v|^2$,

$$I_T^{N|X}(X^{(T)}) \leq 4 \sum_{j \in J_T} |V_j|^2. \quad (5.6)$$

We next identify $\dot{L}_T$. By (5.3) and Lemma 3(ii), applied to $a_\theta = R_\theta$ (with values in $[m, M]$) and to h , which is analytic on the right half-plane,

$$\dot{L}_T \sim \mathcal{C}_T(\partial_\theta h(R_\theta)) = \mathcal{C}_T(\partial_\theta \sqrt{H}). \quad (5.7)$$

Since $T_T(S) \sim \mathcal{C}_T(S)$ and $\dot{L}_T$ is real, Lemma 2 gives

$$\dot{L}_T T_T(S) \dot{L}_T^\top \sim \mathcal{C}_T(q), \quad q = S (\partial_\theta \sqrt{H})^2 = \frac{1}{4} R (\partial_\theta \log H)^2. \quad (5.8)$$

Let P_T^+ be the diagonal projection in Fourier coordinates onto J_T . Taking expectations in (5.6),

$$\mathbb{E} I_T^{N|X}(X^{(T)}) \leq 4 \text{tr}\{P_T^+ F_T \dot{L}_T T_T(S) \dot{L}_T^\top F_T^*\}.$$

By (5.8), unitary invariance and (3.2) with $P = P_T^+$, after division by T the right-hand side differs by $o(1)$ from

$$\frac{4}{T} \text{tr}\{P_T^+ F_T \mathcal{C}_T(q) F_T^*\} = \frac{4}{T} \sum_{j \in J_T} q(\omega_j) \longrightarrow \frac{2}{\pi} \int_0^\pi q(\omega) d\omega,$$

where the limit is (3.3). Therefore

$$\limsup_{T \rightarrow \infty} \frac{\mathbb{E} I_T^{N|X}(X^{(T)})}{T} \leq \frac{1}{2\pi} \int_0^\pi R(\omega) [\partial_\theta \log H(\omega)]^2 d\omega. \quad (5.9)$$

Combining (5.4), (5.5), and (5.9),

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \frac{1}{2\pi} \int_0^\pi \{(\partial_\theta \log S)^2 + R(\partial_\theta \log H)^2\} d\omega. \quad (5.10)$$

For the choice (5.1),

$$\partial_\theta \log S = \frac{\dot{R}}{1+R}, \quad \partial_\theta \log H = \frac{\dot{R}}{R(1+R)},$$

hence pointwise

$$(\partial_\theta \log S)^2 + R(\partial_\theta \log H)^2 = \frac{\dot{R}^2}{(1+R)^2} \left(1 + \frac{1}{R}\right) = \frac{\dot{R}^2}{R(1+R)}, \quad (5.11)$$

and (5.10) becomes

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \mathcal{I}. \quad (5.12)$$

□

Proof of Theorem 1. Combine Propositions 1 and 2. □

VI. WHY THE AUGMENTATION IS OPTIMAL

The particular choice (5.1) is not ad hoc.

Definition 1 (Admissible factorization). A family $(S_\theta)_{\theta \in \Theta}$ is an *admissible factorization* if it satisfies the hypotheses of Lemma A.2: each S_θ is real and even, $\theta \mapsto S_\theta$ is C^1 from Θ into $A(\mathbb{T})$, and $s_- \leq S_\theta(\omega) \leq s_+$ for constants $0 < s_- \leq s_+ < \infty$. We then put $H_\theta = R_\theta/S_\theta$ and define $X^{(T)}$ and $L_T(\theta)$ by (5.2).

For an admissible factorization, $L_T(\theta)T_T(S_\theta)L_T(\theta)^\top = T_T(R_\theta)$ still holds, and Lemma A.2 gives (5.4). The matrices $T_T(R_\theta)$ and $T_T(S_\theta)$ need not commute, so (5.3) is not available; instead, differentiating (5.2) and using both parts of Lemma 3 (with $f(x) = x^{1/2}$ and $f(x) = x^{-1/2}$) together with Lemma 2,

$$\dot{L}_T \sim \mathcal{C}_T((\partial_\theta \sqrt{R}) S^{-1/2} + \sqrt{R} \partial_\theta S^{-1/2}) = \mathcal{C}_T(\partial_\theta \sqrt{H}).$$

The rest of the proof of Proposition 2 applies verbatim, with $1+m$ replaced by s_- , up to (5.10). Since $\partial_\theta \log S = \partial_\theta \log R - \partial_\theta \log H$, this reads

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \frac{1}{2\pi} \int_0^\pi \{(\partial_\theta \log R - \partial_\theta \log H)^2 + R(\partial_\theta \log H)^2\} d\omega.$$

Fix ω and write $a = \partial_\theta \log R_\theta(\omega)|_{\theta_0}$ and $u = \partial_\theta \log H_\theta(\omega)|_{\theta_0}$. The integrand

$$(a - u)^2 + Ru^2$$

is strictly convex in u , with unique minimizer $u^* = a/(1+R)$ and minimum value $a^2 R/(1+R) = \dot{R}^2/[R(1+R)]$. Uniqueness refers to the value of $\partial_\theta \log H$ at θ_0 , not to the family: every admissible factorization with $\partial_\theta \log H_\theta|_{\theta_0} = \dot{R}/[R(1+R)]$ attains the minimum. For instance, let $c \in A(\mathbb{T})$ be positive, even, and independent of θ , and set $S_\theta = (1+R_\theta)/c$. Since c is continuous and strictly positive on the compact torus, it is bounded above and away from zero; by the Wiener–Lévy theorem, $1/c \in A(\mathbb{T})$. Hence this is an admissible factorization, with $H_\theta = c R_\theta/(1+R_\theta)$. Taking $c \equiv 1$ gives (5.1). Thus, within this class of augmentations, (5.1) gives the best bound, and that bound meets the moment lower bound exactly.

a. A one-frequency explanation. Why the two bounds meet can be seen at a single frequency. Heuristically, for $j \in J_T$ the coefficients $\tilde{X}_j = (F_T X^{(T)})_j$ are approximately independent circular complex Gaussians with $\mathbb{E}|\tilde{X}_j|^2 \approx S(\omega_j)$, and $L_T \sim \mathcal{C}_T(\sqrt{H})$ (Lemmas 3(i) and 2) gives $Y_{T,j} \approx \sqrt{H(\omega_j)} \tilde{X}_j$. A single frequency is thus described by

$$E \sim \text{Exp with mean } S, \quad N | E \sim \text{Pois}(HE), \quad HS = R.$$

Since HE is exponential with mean R , N is geometric, $P(N = n) = R^n/(1+R)^{n+1}$, with Fisher information $\dot{R}^2/[R(1+R)]$, whatever the factorization. The joint score of (E, N) is

$$\partial_\theta \log S \left(\frac{E}{S} - 1 \right) + (N - HE) \partial_\theta \log H,$$

and its variance $(\partial_\theta \log S)^2 + R(\partial_\theta \log H)^2$ is the integrand of (5.10). For the choice (5.1) the terms involving E cancel, and the joint score reduces to

$$\frac{\dot{R}(N - R)}{R(1+R)} = w(N - R),$$

a function of N alone. This is the score of the geometric law, and it is the centred summand of Q_T . At the level of this limiting one-frequency heuristic, conditioning on N therefore loses no information. This explains the pointwise identity (5.11), while $Q_T - \mathbb{E}Q_T$ is the corresponding sum of per-frequency score-like terms in the finite- T lower-bound construction.

VII. REMARKS ON ASSUMPTIONS

Remark 1. The Wiener condition $R_\theta \in A(\mathbb{T})$ is equivalent to absolute summability of the covariance sequence, $\sum_k |C_\theta(k)| < \infty$. The $C^1(A)$ assumption additionally gives $\sum_k |\partial_\theta C_\theta(k)| < \infty$ for every θ , and more precisely, for every compact $K \Subset \Theta$,

$$\sup_{\theta \in K} \sum_k |\partial_\theta C_\theta(k)| < \infty.$$

No weighted condition such as $\sum_k |k| |C_\theta(k)| < \infty$ is used in the proof.

Remark 2. Strict positivity is used for inverse and inverse-square-root matrix functional calculus and to keep the limiting integrand nonsingular. The theorem can plausibly be extended to spectra that touch zero, but that requires a separate treatment of singular frequencies and is not covered here.

Remark 3. The exclusion of the zero and Nyquist frequencies affects only $O(1)$ observations and is natural for a real process: positive- and negative-frequency DFT coefficients are complex conjugates. It is also what makes the pseudo-covariance circulant vanish on the positive-frequency block in the lower-bound argument.

Appendix A: Finite-dimensional likelihood regularity

Fix $T \geq 3$ throughout this appendix. We verify the score identities behind (4.3) and (5.4) directly from (A1)–(A2).

Write $\Gamma_\theta = T_T(R_\theta)$. By (3.1) and (A2), $mI \leq \Gamma_\theta \leq MI$, and since T_T is linear with $\|T_T(b)\|_{\text{op}} \leq \|b\|_A$, (A1) implies that $\theta \mapsto \Gamma_\theta$ is C^1 in operator norm. Hence the $N(0, \Gamma_\theta)$ density ϕ_θ is C^1 in θ , with score

$$U_{Z,\theta}(z) = \frac{1}{2} \{z^\top \Gamma_\theta^{-1} \dot{\Gamma}_\theta \Gamma_\theta^{-1} z - \text{tr}(\Gamma_\theta^{-1} \dot{\Gamma}_\theta)\}, \quad \dot{\Gamma}_\theta = \partial_\theta \Gamma_\theta, \quad (\text{A.1})$$

and on every compact subinterval $\Theta_1 \subset \Theta$,

$$\phi_\theta(z) + |\partial_\theta \phi_\theta(z)| \leq C(1 + \|z\|^2) e^{-\|z\|^2/(2M)}, \quad \theta \in \Theta_1, \quad (\text{A.2})$$

for a constant C depending on Θ_1 and T . Consequently, $\theta \mapsto \int g \phi_\theta$ may be differentiated under the integral sign whenever $|g|$ is bounded by a polynomial.

Let $\lambda_j(z) = |(F_T z)_j|^2$. For $j \in J_T$ the zero set of λ_j is $\{z : \sum_t z_t \cos(\omega_j t) = \sum_t z_t \sin(\omega_j t) = 0\}$. Since $0 < \omega_j < \pi$, the vectors $(\cos \omega_j t)_{t=1}^T$ and $(\sin \omega_j t)_{t=1}^T$ are linearly independent, so this zero set is a linear subspace of codimension two and hence Lebesgue-null. The same holds for $x \mapsto \lambda_j(Lx)$ with any invertible L .

Lemma A.1 (Original experiment). *Let $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For every $\theta \in \Theta$ and $n \in \mathbb{N}_0^{J_T}$, the probability $p_\theta(n) = P_\theta(N_T = n)$ is positive and differentiable in θ , and the score $U_{N,\theta}(n) = \partial_\theta \log p_\theta(n)$ satisfies*

$$U_{N,\theta}(N_T) = \mathbb{E}_\theta[U_{Z,\theta}(Z^{(T)}) \mid N_T]. \quad (\text{A.3})$$

In particular $\mathbb{E}_\theta U_{N,\theta} = 0$ and $I_T(\theta) = \mathbb{E}_\theta U_{N,\theta}^2 \leq \mathbb{E}_\theta U_{Z,\theta}^2 < \infty$. Moreover,

$$\partial_\theta \mathbb{E}_\theta Q_T = \mathbb{E}_\theta[(Q_T - \mathbb{E}_\theta Q_T) U_{N,\theta}]. \quad (\text{A.4})$$

Proof. The conditional law of N_T given $Z^{(T)} = z$ is

$$K_T(n \mid z) = \prod_{j \in J_T} e^{-\lambda_j(z)} \frac{\lambda_j(z)^{n_j}}{n_j!} \in [0, 1],$$

which does not depend on θ and is positive off a null set. As $\phi_\theta > 0$, $p_\theta(n) = \int K_T(n \mid z) \phi_\theta(z) dz > 0$, and by (A.2)

$$\dot{p}_\theta(n) = \int K_T(n \mid z) \phi_\theta(z) U_{Z,\theta}(z) dz.$$

Dividing by $p_\theta(n)$ gives (A.3). Hence $\mathbb{E}_\theta U_{N,\theta} = \mathbb{E}_\theta U_{Z,\theta} = 0$, and conditional Jensen gives the bound on $I_T(\theta)$.

For (A.4), the function $g(z) = \mathbb{E}(Q_T \mid Z^{(T)} = z) = \sum_{j \in J_T} w_j \lambda_j(z)$ is a quadratic form which does not depend on θ . By (A.2) and the tower property,

$$\partial_\theta \mathbb{E}_\theta Q_T = \partial_\theta \int g \phi_\theta = \int g \phi_\theta U_{Z,\theta} = \mathbb{E}_\theta[Q_T U_{Z,\theta}].$$

All expectations here are finite by the Cauchy–Schwarz inequality, because $\mathbb{E}_\theta Q_T^2 < \infty$ (conditional Poisson moments are polynomials in the λ_j) and $U_{Z,\theta} \in L^2$. Conditioning on N_T and using (A.3) gives $\mathbb{E}_\theta[Q_T U_{Z,\theta}] = \mathbb{E}_\theta[Q_T U_{N,\theta}]$, and (A.4) follows from $\mathbb{E}_\theta U_{N,\theta} = 0$. $\square$

Lemma A.2 (Augmented experiment). *Let $(S_\theta)_{\theta \in \Theta}$ be real, even functions such that $\theta \mapsto S_\theta$ is C^1 from Θ into $A(\mathbb{T})$ and $s_- \leq S_\theta(\omega) \leq s_+$ for constants $0 < s_- \leq s_+ < \infty$; for example (5.1), with $s_- = 1 + m$ and $s_+ = 1 + M$. Let $X = X^{(T)} \sim N(0, T_T(S_\theta))$ and, conditional on $X = x$, let the $N_{T,j}$, $j \in J_T$, be independent with*

$$N_{T,j} \mid X = x \sim \text{Pois}(\lambda_j(\theta, x)), \quad \lambda_j(\theta, x) = |(F_T L_T(\theta) x)_j|^2,$$

where $L_T(\theta)$ is given by (5.2). Then for every $\theta \in \Theta$ the law of N_T is the original one, and

$$I_T(\theta_0) \leq I_T^X + \mathbb{E}_{\theta_0} I_T^{N|X}(X).$$

Here

$$I_T^X = \frac{1}{2} \text{tr}\{T_T(S)^{-1} T_T(\dot{S}) T_T(S)^{-1} T_T(\dot{S})\}, \quad I_T^{N|X}(x) = \sum_{j \in J_T} \frac{\dot{\lambda}_j(x)^2}{\lambda_j(x)}$$

are the Fisher information of X and the conditional Fisher information of N_T given $X = x$, both at θ_0 , with $\lambda_j(x) = \lambda_j(\theta_0, x)$ and $\dot{\lambda}_j(x) = \partial_\theta \lambda_j(\theta, x)|_{\theta_0}$.

Proof. Law of N_T . Both square roots in (5.2) are real symmetric, so $L_T(\theta) T_T(S_\theta) L_T(\theta)^\top = T_T(R_\theta)$. Hence $L_T(\theta) X \sim N(0, T_T(R_\theta))$, and the $\lambda_j(\theta, X)$ are the intensities in (2.1) computed from $Z^{(T)} = L_T(\theta) X$.

Smoothness. By (3.1), $s_- I \leq T_T(S_\theta) \leq s_+ I$. By Lemma 3(ii) at fixed T , the maps $\theta \mapsto T_T(R_\theta)^{1/2}$ and $\theta \mapsto T_T(S_\theta)^{-1/2}$ are C^1 in operator norm, hence so is the invertible matrix $L_T(\theta)$. Consequently $\lambda_j(\theta, x)$ is a real quadratic form in x with C^1 coefficients, and on a compact neighbourhood $\Theta_1 \subset \Theta$ of θ_0 ,

$$\lambda_j(\theta, x) + |\partial_\theta \lambda_j(\theta, x)| \leq C \|x\|^2, \quad \theta \in \Theta_1.$$

Differentiation of the marginal. The joint law of (X, N_T) has density $\pi_\theta(x, n) = \psi_\theta(x) K_\theta(n \mid x)$ with respect to Lebesgue measure times counting measure, where ψ_θ is the $N(0, T_T(S_\theta))$ density and

$$K_\theta(n \mid x) = \prod_{j \in J_T} k_{n_j}(\lambda_j(\theta, x)), \quad k_\nu(\lambda) = e^{-\lambda} \frac{\lambda^\nu}{\nu!}.$$

For $\nu \geq 1$, $k'_\nu(\lambda) = k_{\nu-1}(\lambda) - k_\nu(\lambda)$, while $k'_0(\lambda) = -k_0(\lambda)$. Since $0 \leq k_\nu \leq 1$, it follows that $|k'_\nu| \leq 2$ on $[0, \infty)$. Therefore $|\partial_\theta K_\theta(n \mid x)| \leq 2 \sum_{j \in J_T} |\partial_\theta \lambda_j(\theta, x)| \leq C \|x\|^2$ for every fixed n . Together with the analogue of (A.2) for ψ_θ (with M replaced by s_+), this bounds $|\partial_\theta \pi_\theta(x, n)|$, uniformly on Θ_1 , by $C(1 + \|x\|^2) e^{-\|x\|^2/(2s_+)}$. Hence, for each fixed n , $p_\theta(n) = \int \pi_\theta(x, n) dx$ may be differentiated under the integral sign. No interchange of differentiation with the infinite sum over n is used or needed here.

Scores. For Lebesgue-a.e. x , all $\lambda_j(\theta_0, x)$, $j \in J_T$, are positive, because $L_T(\theta_0)$ is invertible. At such x ,

$$\partial_\theta \log \pi_\theta(x, n)|_{\theta_0} = U_X(x) + U_{N|X}(x, n), \quad U_{N|X}(x, n) = \sum_{j \in J_T} (n_j - \lambda_j(x)) \frac{\dot{\lambda}_j(x)}{\lambda_j(x)},$$

where $U_X = \partial_\theta \log \psi_\theta|_{\theta_0}$. The previous step therefore gives

$$\dot{p}_{\theta_0}(n) = \int \pi_{\theta_0}(x, n) \{U_X(x) + U_{N|X}(x, n)\} dx,$$

that is,

$$U_{N,\theta_0}(N_T) = \mathbb{E}_{\theta_0}[U_X(X) + U_{N|X}(X, N_T) \mid N_T].$$

Information. Conditional on $X = x$, $U_{N|X}$ has mean zero and second moment $I_T^{N|X}(x)$. With $Y = F_T L_T(\theta_0)X$ and $V = F_T \dot{L}_T X$ we have $\dot{\lambda}_j = 2 \operatorname{Re}(\bar{Y}_j V_j)$, so $\dot{\lambda}_j^2 / \lambda_j \leq 4|V_j|^2$ and $\mathbb{E} I_T^{N|X}(X) < \infty$. Hence $U_X + U_{N|X} \in L^2$, the cross term vanishes, and $\mathbb{E}(U_X + U_{N|X})^2 = I_T^X + \mathbb{E} I_T^{N|X}(X)$. Conditional Jensen applied to the previous display completes the proof. $\square$

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